ActInf Livestream #021.2 ~ John Boik

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hello everyone welcome thanks for joining actin flab this is live stream number 21.2 i think and this is our sixth discussion with john the first four were background discussions where john and i worked through these jam boards and through the three-part series that john had written science-driven societal transformation and now here we are today on may 11th 2021 in our second participatory group discussion so this is going to be an unpacking and an extension and an opportunity to share perspectives and ask john questions so thanks to all who have joined live and who will be joining live and also if you're watching on youtube feel free to leave a question in the chat and we will relay that to you so we're just going to go around with a quick round of introductions maybe each of us could just say hello and something that they're thinking of or something that they're excited to talk about today and then we will pass back to john and just start off on this road together so i'm daniel and i'm in california i will pass to lee hi uh i'm lee i'm in i'm in london at the minute um although i'm a phd researcher at the university of york uh john i'm today i'm interested to talk about um maybe a minimum viable prototype for one of the clubs um and also uh a little bit around the the metaphor of the super organism a bit as well if we have jan uh and our path to uh scott scott i think thank you my name is scott david i'm at the university of washington applied physics lab where i'm the director of information risk and synthetic intelligence research initiative delighted to be here today and i'd also i'm very very interested in chatting about the scale scale independent variables how the interventions at one scale can help to alter system performances at other scales due to fractal nature of some of the affordances we're talking about great interest to me and i'll pass it to dave douglas i'm in the philippines 120 miles north of manila where it's cool because it's a mile high city and i just came across the answer to a question what is a state in the parcels and particles paper dr fristen says a state is a particle so yes it really really is monads not only all the way up but all the way down to and uh who's left daniel you've already we already know you we'll go to we'll go to john thanks for the the spicy introduction dave i'm john boyk um if you go to set two slide one uh this this series this these numerous talks are on the topic of that's the title of a three-part series that i recently published in the journal sustainability um part one is world view part two is motivation and strategy and part three and three is design i i i'm looking at societal transformation through a particular lens in this series and that lens is the de novo development of fundamentally new systems so when i say systems i'm talking about the economic systems governance systems legal systems et cetera the we're going to focus today i think on maybe some of the practical aspects like how would this be done and uh i would like to just start by saying just a few words about this the kind of theory of change that i talk about especially in paper number two if i were brilliant which i'm not and i came up with this amazing new design for societal systems new governance systems etc if there were no way to actually practically implement that then i it would have just been an academic exercise so part of this series is arguing that there actually is a reasonable way to move forward a a viable path to move forward that is doable that that actually could be put into into implementation so this the series is is itself is is essentially um outlining an r d program for the development of this framework to study and uh create and develop and test new integrated systems and that happens at according to this theory of change that happens at the at the local level so the concept is is that this r d program is a is a collaboration between the global science community and local communities and the new systems are implemented at the local level on a volunteer basis in what i call a club model so um the the concept there is that the scientific community would have developed the tools and technologies and resources things necessary to make these new systems work well uh and um they the local communities some volunteer local communities somewhere in the world would would look at the results of all the computer simulations and all the studies that have been done and say that looks interesting gee it looks like our community might really benefit from this um uh let's give it a field trial let's give it a try and uh so that's the partnership and then the community itself now this could be i don't mean a city i don't mean a nation i really mean like perhaps a subset of a city so say just a thousand people or something like that it might be enough to start a club and the club would function everyone in the club would function as normal in society except that on top of on top of you know there's another layer of organization on top of the normal layer of organization and that's this club level so civic club you can you can think of it as a civic one and the civic club has all these you know uh ways that they go about decision making and exchanging currencies and various things that allow them then to test out these new frameworks of of uh societal systems so that's the gist of it and i just want to say a few things about this club model before we really get going just so we've kind of all on the same page on that um just before i do that maybe go to slide two of that same deck i'll point people to uh principled societies project dot org if they're looking for more information on all of our videos are posted there and the papers are posted there and there's some other materials posted there so for for information go to psp and let's go to slide 17. okay oh that is not the right uh not quite the right one i wanted the theory of change one and that's not it [Music] uh which slide is that you're looking at seventeen one yep there you change you know that's somehow i'm uh oh oh here we go yes there is change and strategy uh uh this is just kind of uh capturing what i just said the idea is to implement new fundamentally new systems at the club level uh i'll call the club a community but this community could be relatively small than just a portion of the city and the reason i'm one of the reasons i'm looking at the club level is because i see it as kind of the goldilocks level it's the it's the smallest size organization of a community that could make the club able to replicate right so and and so if the if we were instead just looking at say an individual business that business like i say say i was promoting a cooperative business model or something like that that business would have to function within the wider set of businesses and would have to compete in the normal market for the consumer's attention and uh it would have a hard time doing that because it's operating according to some different set of rules in a normal business so the at the club level i see that as just large enough that all of these new functional systems can be tested and and it's large enough that the the organization the community can be self-replicating the club model can be self-replicating it can grow in a local area and the idea can spread to other local areas um uh all right uh slide 19 please can we just go to just lee and then scott just because we're getting hands raised on each of these slides so yeah so lee and then scott sure john what are the um in order to start a club what are the minimum requirements that are needed do they need any funding or are there other set of conditions that it might need um ideally there wouldn't be any requirements ideally there would be support from the from the you know the global community to start a club but but but the but the clubs are designed such or at least later we'll today we'll talk about a simulation i did of a of a club model a particular framework i called the lida framework but i designed that so that it could bootstrap its own growth so there was no it was not necessary to you know for it for uh to reason to have millions of dollars say to start the club or anything like that it really just grew step by step exponentially fast so in the first few years hardly anything was happening and then it started to grow you know very rapidly after that uh so you know what would be necessary mostly just the individuals that are that look at the by the time a a field trial would ever be started there would have been you know let's say hundreds of papers scientific papers and simulations and all other aspects of this so everyone would know what they're getting into this would be very clear what the what the you know what the new systems are and what they do and how they and then at that point all that would really be necessary is for some community somewhere in the world to say oh that looks really interesting let's do a field trial you know here's we have a thousand people in our little in our community that are interested in this and let's give it a shot so that's the main thing i mean maybe a little bit of funding would be necessary but not an extreme amount so it sounds oh it sounds almost like there's a prelude phase which is the what we're in and then the stage will be set for proper emergence for a community to then take that next step but um it's a really interesting question about the viable system so i hope that we pursue that line as well yeah john anything else and i did i just maybe we'll add that by the time the first field trial ever starts there really should be no surprises about any of this i mean the community would you know conduct the trial but it should be a foregone conclusion what's going to happen as far as you know how it works and what the benefits are because this should have been worked out very well already so uh you know it shouldn't be as it's not like uh here's a you know here here's a new here's a new machine a tractor or something now see if you can drive it it's not quite like that because there would be uh you know training involved we would realize what the purpose of the tractor is what we're going to accomplish with the tractor how it works how you repair it you know like it would be a uh again there's just should be very few surprises by the time the first field trial starts that doesn't mean a community wouldn't be able to make decisions on how it what it funds and you know how it functions of course the community would make decisions about how about its the kinds of jobs it creates and things like that but the fundamental mechanics mechanics of the system should be pretty well worked out just very quickly then so is the um is the particular community so you've got this prelude phase where you're kind of working out all the details of the de novo design is that done with the community oh partly i mean that would be with any communities like globally in a sense that prelude part is the global science community and any individuals or communities anywhere in the world that want to participate provide feedback do user testing um you know all the little parts of the systems would be also tested in various groups and there would be like really just a ton of testing that would occur before the before the first formal field trial ever started um and the and that particular community what could have input on um you know what kind of particular design elements it might want in its system so and and that's actually a good point that i'm glad you said that because i also wanted to say that the purpose of this r d program is not to build one system that fits all it's to build it's it's an ongoing exploration of them you know different pieces and approaches and ways you might create systems such that they fulfill their purpose so there could be a wide variety of you know parts to choose from or ways to do things or you know hopefully there would be numerous kinds and types and pieces of systems that would be tested over time that communities could you know implement then and all with the purpose of of uh you know fit for purpose you know all with it with the concept of fit fit for purpose behind us got and then anyone else who raises their hands i just wanted to go back you were talking a little bit about the size of groups in the clubs it's a the optimization's going to be really interesting because you might have and dunbar's number involved there that dumbar's number concept being the most people you can have as your friend kind of idea so whatever is 120 or something but one of the things that's interesting also you have the internal voice dynamics of the club plus the external voice dynamics of the club so as the club coheres an internal voice it will have a more coherent external voice which will allow it to entrain and harmonically couple other systems more effectively right because it'll be more clear here and one of the things that i'd like to get into later in the conversation is we have uh some ideas we've been working on on scaling how legislative judicial and executive function the three functions of all governance systems have rule making enforcement of the rules and operation of the rules how those naturally layer on just as a function of scale as you increase in scale from rule making you have to go into different abstractions for enforcement and operation under those rules and so that would be very interesting to look at the dynamics of club growth as an internal process of internal voice and a recognition of the layering on of governance that occurs internally and then how that governance needs to expand into the external environment in order to be more effective the the bottom line is if you have a system a club where we say we are the free speech club we have the rights to free speech and there's no other club that has a duty to respect that then you don't have a right to free speech outside of the right and say how do you iterate the story outside of the club so that it becomes a reality that's shared so it can be enjoyed right right i'd like to talk about that yeah yeah there's there's a there's a ton there that we go through uh i have these concepts in mind as i was designing these systems or designing the frameworks so all good points um just a little bit on size i'll try to talk about a couple of things briefly that you raised a little bit on size um so so in our brains we have a community of neurons that are that are functioning to help the body make decisions and that that community of neurons is very much connected with the rest of the body also feet and the liver and the heart and everything else well how does all of that you know how does how do all those cells and tissues and organisms cooperate together to make the body function the way the body functions one thing that happens is there's a very rich stream of communication that is passing between all these different cells and tissues and things for hormonal signals and small molecule signals and blood is circulating and there's all these this awash in signals and communication so uh communication is one of the prerequisites for any kind of large organized you know instead of individuals whether they're cells or whether they're people to function together so one of the questions that we can talk about today um is what kinds of tools are necessary to allow for rich communication to occur in say a group of let's just say 10 000 people or a hundred thousand people or a million people how do how does a large group of people come together richly communicate their narrative their their their both you know i want to tell i want to tell my group what is happening for me what i anticipate will happen if we choose a over b over c uh where this might lead in the short run where this might lead in the long run how this fits into other issues issues d and e and f i mean you know i'm an intelligent person everyone else has intelligence and um that intelligence can be used in the group uh cognitive you know the group cognition in order for the group to make wise choices about you know actions and and where they want to put their attention so so the tool question is a big one and i would say that that currently there are no tools that allow a large group of individuals to come together to collaborate with via rich communication and rich storytelling right we that doesn't exist yet i would say and i have ideas on how we might create that but that's one of the that's one of the one of the tasks that would have to happen over the over the first several years of the program is addressing that particular problem of how does a group richly communicate and collaborate and making decisions so so if we're just talking to get to your point now if we're if we just have a small group and we're just just talking we're working out things we're making decisions together well maybe that group can't be more than a few people before it starts to fall apart or you know maybe it can't be more than 10 or 20 or 100 people or something um but if we have some way to richly communicate then maybe we can scale that you know we can scale that communication uh to a larger group to thousands or ten thousands and still there might be some limit be beyond what a club wouldn't you know like maybe a club should never be more than a hundred thousand or never more than a million or never more than ten thousand you know who knows and so that's between that's of individuals in a club and then there should be similar communication between clubs that must happen and that's getting to a that's touching another point that you raised is how do how does a club reach reach the you know interact with the wider world the the both and and the my framing both with other clubs how how does decision making occur between clubs how does communication happen between clubs and then also how does communication happen with the rest of the world thanks john um stephen and then scott and then anyone else who raises their hand yeah thanks i i think i'm interested by this um the word purpose you know we think about purpose and all of this and clubs tend to have a particular uh focus or purpose as you're mentioning i think it's interesting you're bringing in the science intentionality you know the intentionality of the science community to enable sort of seeding so i'm wondering how that might you might think about that sitting alongside say community organizations or like community hubs for instance that or multi-service agencies in the community or nodes of support those types of things this might be something which could fit in there and give this kind of club purpose that might complement or is it different similar to some of those types of service provisions that exist out in towns and cities i'm not exactly sure the question you're pointing to but let me say a few things and maybe i'm going to touch on it um so so um kind of in the abstract the the you know the concept behind this whole program is that um that that an organism has an intrinsic purpose and that trans intrinsic purpose is to thrive both now in the future so expected thriving and uh in order to do that it uses cognition right and via cognition and we've talked about how active inference fits in to that whole picture via cognition it really reduces the uncertainty about its capacity to thrive it's the essential variables that it you know that it that it needs that it functions the world it functions in and whether it's need true needs will be met right so uh that's an organism and i'm viewing a society as an organism right so in the sort of the larger picture the abstract little abstract the question is you know is is a community functioning in a way is a new system functioning in a way that is reducing uncertainty about the capacity to thrive so that's that's uh that would be yeah that would be you know uh implemented in a way via some kind of fitness metrics so just like in normal society in a regular society we have say gdp and other various types of you know information we gather and metrics that we use there would be a set of metrics that would go along with this whole program so that a community would know how it's doing and how it will how it can expect to be doing two years from now and five years from now and ten years from now you know maybe even 20 years from now so so there may be many subgroups within say that say that there's a club and the club is 10 000 people for example clearly there would be some groups within that club that would be interested in various aspects of you know topics or ways of doing things or various kinds of communication and you know maybe some some subgroups are really focused on the social components of the community components the you know maybe emotional components social components and then other groups are interested in technical components and modeling or whatever you know other other groups are interested in the environmental components for example well that's great that's you know association between groups is a part of cognition our neurons associate in in ways that allows for cognition of the whole and in the same way a you know club in a community would do the same so there could be all kinds of subgroups that are all functioning but the umbrella purpose would be present right because it would be reported daily say or even hourly or weekly you know there would be everyone would know how how we're doing as a whole and that we're all in this together to raise our our joint capacity to thrive and and maintain our maintain our ability to thrive so yes there would be all kinds of groups that would focus on different aspects of this and the umbrella you know reporting and collecting of information would be happening still across groups across the community so that there would be some sense of how are we doing and what's going to happen next you can i just reply slightly to that yeah just very good yeah just that's really interesting and it sounds as well this could tie in with the kind of another type of evaluative capacity or ongoing rolling evaluative capacity but with an agent based or active inference based engine so to speak so that it's giving update to communities around how they are progressing so to speak when that may help them in other areas some of which may be outside of the club's more bounded area you know it gets into the niche and complexity so the interesting if you thought about this might be how this might link in with some of the new developments in evaluative evaluation going on and that might be quite a good avenue yeah there would be i mean if you were going to collect information uh and and assess how we're doing and what's going to happen next there would be all kinds of evaluation that would have to occur not just on the simple things like you could think of some simple things right off the bat like do we have enough food do we have enough water well fine but humans are complex organisms we have complex needs social needs you know emotional needs you know like physical needs you know we and all of those needs should be taken into account in in some kind of you know comprehensive evaluation of how are we doing and what's going to happen next and and obviously every any organism has to know that you know every organism wants to know it needs to assess its status like how am i how am i doing am i too cold am i too hot am i hap unhappy am i happy you know like every organism needs to do that and a community a community as an cool thanks john also welcome blue and on the the goldilocks note and this narrative note i really like what scott had said about the internal and external communications and as internal communications become structured from a connections perspective but also from a narrative perspective the external interfaces and narratives can be structured so goldilocks doesn't just evoke the um sort of upside-down u-curve i mean this is something that we know from pharmacology from temperature everything there's too much and there's too little and there's a failure mode on either side just like a bowling alley and of course the middle solution is going to be better it just wears the middle but goldilocks also reminds us that that middle solution has to be more like a child's story rather than some sort of ab truce optimization problem so it's kind of a fun reference that it should be comprehensible internally and also externally in order to be functional right right right so yeah and it may be just to touch on that you know from this to say the scientific viewpoint one might think about fitness and you know in terms of metrics and statistics and things like that but other people won't relate maybe so much to that kind of assessment so everyone who's involved has to there has to be a way to communicate the ideas of how are we doing and and you know what is our status in ways that are meaningful to everyone involved or else if it's not meaningful will really won't work right so so some some people might might discuss uh you know status in very technical terms and other people might discuss status and more say emotional terms or social terms or something like that but but the general sense the general the general purpose has to make sense to everyone or else why be a part of this one other thing on the goldilocks is the question is um to be clear it's not my intent that a club say a club started in in houston texas with a thousand people it's not my intent that the club would stay at a thousand people it would grow if the whole concept of this is that if this new model this new club model these new frameworks actually work if they actually empower a community to do better to to have be more secure to have a higher quality of life to have economic security to you know in numerous ways if these clubs benefit the population the way i think it would then they would grow because it's kind of like the you know why don't we use typewriters anymore because we have computers that work so much better i mean they're just so much better who would want to use a typewriter you know that kind of thing so which that's the concept of this it should be so easy to join a club it should be so easy to participate that that like why not my life will be better if i participate in a club and if that's if that comes true if it really works as advertised then clubs will grow and the club model will spread to other places and then instead of having a club of ten thousand or one thousand you have a club of a hundred thousand and every city has a club and pretty soon the you know much of the world is starting to function along these you know with these with this along with these new systems much of the world is engaging with them or enough of the world is engaging with them that the ideas behind them suddenly become the start to become the dominant ideas so that the goal is not to help a community so much the goal is to actually transform globally society or at least semi globally thanks john i'm gonna ask a question from the chat and then stephen and lee so in the chat natalia writes i read in the paper thanks for reading the paper natalia that people would keep their income when not working i'm curious what role motivation plays in the success of the and what kinds of structures are in place to manage that so agent level motivations and how do we design incentive structures and ecosystems for participation that are able to incentivize various kinds of participants all right okay so so what what you know what motivates us today in our economy um you know a variety of things we're complex organisms we're motivated by a variety of things but a large one because the way the system is designed is that if you don't participate if you don't go out there and get some kind of job you're gonna be in the street you know there's this like this kind of you know you're forced to participate and if you don't and if you don't you're going to be in trouble well well that's not very useful uh for many reasons far more useful is to offer people what they actually really want what what humans tend to really want which is a variety of things they want to help each other they want to be part of a community they want to learn they want to develop skills they want to do any number of things that engage them with each other and in they want to solve problems they want to be a part of a problem-solving process they want to they want to you know cooperate it makes us happy to cooperate right all of nature is cooperating like this cooperation is the rule rather than the the exception in nature so how can we build a system that just allows us to really be ourselves to really get what we want most want right like i want to be in a in a kind and cooperative and functional uh setting a group i want to work with each other i want to work with the people around me and and help each other and they want to help me and so how do we make that happen at scale right that's the question and uh so the motivation isn't to get rich i mean we'll talk about the leader framework in a little while and that's just one example of a way to organize an economic system but but that particular system actually achieves uh income equality over time so clearly motivation isn't to make money like that's not why you're engaged in the economic system the economic system is only a it's really a uh it's a cognitive system not a not a money system money is just a tool that's used as a as a way to share information right um so in the in the leader framework incomes equalize over time and the motivation to engage is to be to fulfill all these deeper human needs that we all have thanks for the response john so stephen and then lee yeah i'll just sort of carrying on a bit from what you're just saying there with um the community getting involved their benefits to the purpose and to their motivation at the community level um that is a missing link at the moment in terms of and it probably was much more present even 100 years ago 80 years ago you'd have had the women's institute and you had all sorts of different so we professionalized an awful lot of roles okay and you see that with nursing and all sorts of professions where they've increasingly professionalized and trained so the question i've got there is how do you um bring in the motivations of the professionals who have another niche that they're trying to optimize around maybe writing papers getting a career and things like this and do they have to be humble in some way to honor the input from this community because obviously you're trying to invert some of the assumptions about where everything should be located i.e it doesn't need to all be driven by professions and just keep driving more professions that you can bring in community input and wisdom to also be part of that through these clubs but how do you marry that the types of niches that people are in and the competition for their time and right right um i don't i don't actually see that as too much of a problem let's just let's just pick a profession uh say nursing or something like that and then the question becomes uh let's say that i live in houston and the club has started and why would i want to you know i'm a professional i make a reasonable salary let's say why would i want to participate in a well there might be numerous reasons i might see i might understand the potential of the club i mean you know there's this whole body of literature now these simulations and other literature that suggests that this club model might actually empower the people who belong to a club to make better decisions to achieve you know higher environmental quality and higher social quality so i might be interested uh just to join for that reason because i see it as a beneficial thing for the larger community right i can engage with these people and make decisions and together we can you know make beautiful things happen uh but let's now let's focus just on the economic aspect and again uh you know there what might be any number of ways to design any of these systems economic systems governance systems you know there could be all kinds of concepts that we that the r d program could assess and try and test out right but of the in the leader framework the kind of prototypical economic system that we hopefully will get to talking about here um um the question becomes how do we get from here to there like how do we get from today's society and incomes and structure to some other structure you know down the road right so i try to show that in the simulation that model that i published it goes over a 30-year period so so again the kind of the expansion of the club and its activities is exponential so maybe in the first few years there's not a whole lot happenings people join but you know there's not big effects but then these it bootstraps itself economically and otherwise and then they're starting to be bigger effects so uh you know just kind of jumping to the chase a little bit in the leader framework it's designed so that incomes rise over time right and the the people in the simulation people only join if they're going to be better off economically otherwise there's no reason to join in the simulation so um i i let's say i'm a nurse i have a pretty good income why would i join well if i join at any point i will get a babes like a um and my income is high the let's say the inc the lowest target income is rising but my income is still higher than that i can still join and i'll get a kind of a i'll get a a small a small bonus uh that's not the word i use in the paper but it's kind of like that i i'll get a i'll get a say i don't know what let's just say a thousand dollars a year for joining well that's a thousand dollars a year that i didn't have otherwise so i even i get a even though my income is high relative to where this income target is growing i still get a benefit from it and then i get to use that money and that you know that part of it is a local currency so i get to use the dollar dollar currency the national currency and this local currency in the decision making process and i'm engaged you know i have i have some skin in the game now and then as that uh income rises at some point it's likely to reach mine or go above what my current income is and then for sure i want to join because if i join my incomes you know like my annual income is going to rise maybe substantially so the in the in the simulation the income target rises up to about uh somewhere around 110 the equivalent of about 110 000 per year per family so you know that's more than some nurses make uh and especially for those nurses that are at the bottom level bottom you know economic level or teachers or firemen or police or whatever you know uh there would be motivation to join because your income is going to not only increase but become stable like you don't have to if you lose your job there's no worries you're going to be taken care of this is circulating currency in such a way once you're in you're okay it doesn't matter if you're working or not it it's john it's it's a way to rethink driven by money versus not driven by money because you've really said it both ways you've said people are not motivated by money yet you've repeatedly appealed to the idea that every person will gain more money and implicitly that that's what is important and so it's actually just it is what it is and that's where active inference and all these other techniques and extended cognition come into play because there'll be a variety of narratives for people who are in positions so we'll go to lee and then scott and then i'll ask a question in the chat uh yeah so um it's kind of a related point actually so uh in the paper um i remember you were reading that at one point you're talking about you know effectively as the dinobo system uh starts to out compete the native system you know there are gonna be winners and losers effectively so you know if you're a franchise owner of a of a mcdonald's or something you know your cheap labor disappears so on a larger scale right that means you know they're um well i guess what i'm what i'm i'm asking is do you anticipate that ultimately the current system you know with lots of vested interests to remain the way it is he's going to start pushing back once it realizes that this kind of club model is potentially a threat to it yeah so right so it's true there's winners and losers in this and the and in a sense if if if you're measuring winners and losers by does this civilization survive and thrive well then there's hardly any losers because you know the the whole concept of this is so that societies and civilization can thrive into the future right and that's not and you know just as a counterpoint that's not our current system you know we're our current system is essentially on a suicide uh route you know trajectory of destroying life on the planet so the current system is not working uh and you know it's not working in the sense of our insecurity about will we even be here in a hundred years is is quite high right so uh but but in a in another sense yes there will be losers and winners so the idea is that this r d program is creating and testing new ways of organizing society that really do that really function well and any organism that functions well is going to out compete other organisms you know that are not functioning well right in that sense so um a community say that is using these tools and using these systems and it's really flourishing uh probably we'll be doing flourishing more than other communities that aren't and that's the reason that other communities might say hey that looks good but let's do that now you have a subset of say you know multi-millionaires and billionaires and at least on some perspective they will be losers because uh you know that that's not gonna work that's not gonna be compatible with this new way of doing things uh and you raised one of the reasons why i wouldn't be compatible is that where are they going to get their low-wage workers to work and you know work in their factories and stuff where are they going to come from if all the other folks are having high-wage jobs in these new you know these new systems right these new frameworks so the the business model will start to falter the conventional business model will start to start to falter and as it should because you know that what what are we what are we doing here what is the what are we doing in society right are don't we want to flourish don't we want systems that help us to to reduce our uncertainty and flourish into the future well i i would think we do overall right so systems that don't do that really have to fall by the wayside or or transform themselves into into new systems that do function better and the the long-term aim of this r d program isn't necessarily that the world will be using these kinds of systems another option another you know potential future is that current governance systems current legal systems current financial systems you know uh all of our current systems might learn from these experience these experiments everywhere and go you know what that's that's actually a good idea that works well let's incorporate that into our structure and uh you know we'll be better off everyone will be better off so that would be fun maybe that maybe the clubs grow to a point around the world and they just they the ideas just assimilate out into the larger world that are taken up and then everyone is still doing better so that would be great or maybe it doesn't go that way maybe maybe it becomes a you know like us against them in a sense there's the clubs that are growing the the native systems the current systems are not doing well they refuse to learn from these experience experiments they're they're not interested in change well then they probably will be dinosaurs you know i mean i'm just thinking about uh so there are examples of um a community energy generation for example so those solutions start to exist um you know and then there are examples of invested interests where they've kind of seen that as a potential threat and move to outlaw them so i mean it's just those kind of you know it's it's those kind of conflicts i was wondering where they might yeah you know um of course those kinds of conflicts will happen um and the and the r d program and the people involved and creating this whole system have to think these things through carefully to try to avoid you know problems that might be foreseen right obviously so these are very important questions and we would have to give a lot of thought as to how we you know skirt past these problems now you know and and i you know i don't that's to be done that's work to be done right i have a few ideas and that's and i and i can say that's one of the reasons i wanted the global science community as a kind of a you know co-leader in this whole process it is for one thing it's global right it's not limited to the laws of a particular nation so and it's authoritative in the sense we can write papers and we can do simulations and we can talk with some authority of it would this be a good you know do these things help humans would this be good for human societies and civilization you know we can we can do we can do that we can we're independent of any particular nation in a sense um and then suppose that there's a nation that says no way this is against the law you can't do it here well okay you can't do it there but there will be other nations that wouldn't have those rules and this club model can thrive wherever it's allowed to thrive right and you would think that there would be some nation somewhere in this world where there might be some allowance for these kinds of social experiments to occur and once it works that's the thing that's the concept is that once it's shown to work you know first in one community say and then maybe in two and then four and then eight you know once it's once it looks like this actually works then there's a whole nother slew of of papers and simulations showing you know like working you know assessing that and that information is broadcast to the world and uh you know at some point it becomes harder and harder for a society to to keep it out because you know no no no country uses typewriters anymore you know except the hipster countries so we have except they have except the except the hipsters they'll say i want that good old-fashioned government so scott and then a question from chat and then dave so um great conversation thank you this is so fascinating so the couple of book references and a couple of things i'm going to talk a little bit about money and risk here going back to what we were talking about a little while ago and that moved from money and prosperity you know there's a book by i think it's hannah's men called doing money where the author suggests that money is a universal risk consolidator and so i'll hold up here 20 billion okay okay due to the bearer upon react so it only is as good as the narrative right so um i always carry zimbabwe money around because that's inflation protected in the united states we can't get near the inflation rates of this so um the the reason i raise that is money you know we there's a book called the past as a foreign country where we think people in the past oh they just didn't get it and then we look at this and go you know money actually it's interesting when people want to accumulate money there's a lot of theories of money this smith and wixel and hume but money is a form of communication between people it's an encoding of risk in a sense right when we look at and decoding of risk and a universal risk encoding in a sense and it's kind of interesting because then it becomes abstracted into its own get its own end that's when it comes the commodity of money right and so it's not a medium of communication in exchange anymore but its own gain its own because it's used generically against all risks i can de-risk so many universal risks that's pretty true it's like a universal solvent of water you know it becomes something that's biologically very dynamic and so the idea of the medium of exchange is a fascinating one here so i just wanted to fuss with that a little bit and then so if you look at it that way then the meaning so this is zimbabwe local community meaning and then there's united states local community meeting and they're different because the community is what generates the value not the money this is a piece of paper has exactly the same value as a us dollar but it's the symbol of a system and the system is the community in the communities where the value comes from not from the piece of paper so that's the intrinsic uncovering one of the things that i've been fussing with just to share on this point is so the past people have been doing money and maybe they're not so stupid maybe they've been doing clubs maybe this is a money club right um and so one of the challenges is well now maybe we need a redistribution idea among the clubs and i think one of the things that's really appealing about the way you're describing things is the intrinsic transparency that comes with the exponential increase in interaction volumes so it's impossible to su press transparency among so many interactions is very similar to after 2008 financial break where there were calls to get rid of the rating agencies because that was one stop shop for conflicts of interest you pay them more money you get a better rating and they said instead let's get rid of trading agencies and let's use the spreads in credit default swap markets because it tells you the same information will the company pay its debts as they become to that's what the rating agencies tell you and that's what the spreads and credit default swap markets tell you but the difference is in this in the credit to falsehood markets those are set by thousands of interactions so you can't game them right so that's what distributed governance looks like and that's what intrinsically you're it feels like you're building that from the bottom up but not just vis-a-vis financial instruments but vis-a-vis anything that community as the source of trust and knowledge and then the i call it the community trust community information boomerang because then when you want to authenticate or audit you go back to the community because that's the ultimate source of trust because that's where it came from the first place because it didn't come from the data passing through it it came from the fact of the meaning in the community drenching and enriching the information as it goes through right so i really love the way you're talking i just want to make a few comments on it but that abstraction of money is kind of an interesting thing that you get to levels of abstraction where abstraction itself becomes a form of violence because it's because it reduces the granularity with which a system sees the individual and so just last point kandinsky of the folks on this phone have heard this already but kandinsky made it the painter said once that violence societies yield abstract art and one of the things i always wondered is that reversal is abstraction itself violence and to me it feels like getting to a certain scale it feels like violence try dealing with verizon the other day i was so frustrated i was wanted to slam my phone down on the receiver down but i only had a cell phone i don't have an old style phone so i couldn't slam it down with drama anyway so these are you've raised a whole series of really interesting and points so first what is money you know if we're going to if we're going to design new systems including new economic systems new financial systems new monetary systems what is what is money what is what is what is what is its function in a in a club or in a society right well um already in what i call native systems money is a voting tool if if i have a billion dollars i get to vote for any number of things that i want i can make i can make whole organizations do fulfill my you know desires because i i have enormous voting power now obviously that voting power is not democratic it's not transparent it's you know it's really a dysfunctional uh inflow of information uh and the and quite obviously if we're going to design new systems we would want to it's fine that money is used as an information tool but that information tool should be completely transparent fairly distributed uh you know it should actually function as a way for all of us to share our desires our fears our our our you know our viewpoints our our preferences to the community you know it's redesigned money that serves an informational purpose and is inter and is transparent and is you know deeply democratic in its function that's a you that that would be a useful way to communicate you know uh so so that's part of the task is to how do we do that how do we design a system that functions like how do we design an economic system that functions where money is no longer even viewed as money in the normal way that economics is as a matter of fact that economics the like the economic profession say for a club is not even resembles economics because suddenly we're not talking about economics we're talking about cognition we're talking about a societal cog this is all about decision making under uncertainty in reference to some fitness you know vitality score of like how do this is cognition how do we how do we it's no longer economics now it's a now it's a new field called cognition that happens to use a voting tool called that we might call money a couple things on that so um you know a club can't redesign the same national currency can't redesign the dollar or anything like that so but it can create a currency for its own use within its own within its own milieu that functions exactly the way that i'm describing that that is transparent that is you know does all the things that we've just been describing so i take a shot at that in my design of the lead framework and um if we get to it i'll try to you know show you how kind of conceptually how that works well it's just a quick follow-up that's one of the elements of sovereignty right is the tax and the money issuance the m1 supply you can't use a starbucks card everywhere because that would intrude on the sovereign supply of money on the fiat and so the exertion of authority into every every interaction is has three parties in it because the trading medium is a declaration by the sovereign that they are there to help but it's like um von munchausen by proxy syndrome right they they keep people weak in their interactions in order for themselves to be strong right very interesting just because there's a bunch of questions the chat and a lot of raised hands so i want to get to everyone in order so um dave and then i'm going to ask a question in the chat and then blue and stephen yeah it's not self-evident what the relations of property intellectual property occupation competence uh in the u.s there seems to be a um an insistence that there that the the traditional money relationship structures everything else your competence is determined by who writes your page your credibility is determined by that um it isn't that way every place in mexico um the right to professional association and voluntary association is written into the constitution in the philippines the government is constitutionally obligated not just to tolerate but to actively promote professional associations and voluntary associations uh and reading some of the things and by no means all the things that robert david steele says denmark apparently is doing a fabulous job of managing the knowledge economy and uh moving more and more all the time to open source everything except maybe initiation of government force out and out violence to preserve buildings and children and things like thanks for the comment sure yeah you looked at it steel's uh proposals for um setting up a very much more transparent way of exchanging knowledge and vetting competence no i'm not i'm not familiar let's um share that it sounds interesting though yeah cool and and part of this prelude in the research and development phase is certainly to take stock of various proposals that address different aspects related to transparency accountability information propagation value exchange crypto economics nfts all these things are going to be woven together in a new tapestry that hopefully is our parachute slash trampoline so yvonne yvonne asks a question in the chat he says hello thanks john for the second stream sixth stream it is excellent my question does the club model imply a governance where two or more clubs come into confrontation in their ways of acting so early stage club development or late stage let's just say two clubs don't just believe different things they don't just infer differently but they act differently what do we do sure okay so um so the in the r d program maybe the first task will just be to work out how societal cognition can happen in a club setting you know like well and then all the different systems that support them but very quickly you know once progress is being made towards that end the next big question is how do clubs cognate how how do they communicate so just as there would be just as there would be systems and technologies and frameworks that allow individuals in a club to communicate and cognate together and make decisions together whether they're economic decisions or whether they're governance decisions or whatever kind of legal decisions whatever that is there would be a similar kind of overlay to the to the between club setting right so that those those maybe it would be similar technologies maybe they would be adapted a little bit for the club to club interactions but the same kind of concept holds is that uh clubs the community of clubs is a cognitive community is a you know is a is an organism a cognitive organism so how does it make decisions and view the world and uh and act now it might be that let's say that down the road there are hundreds of clubs or thousands of clubs and one club says you know what we really like this slavery idea that's gonna work really well for us and you know we're gonna we're gonna go off on this tension or you know some some say destructive tangent um well you know i mean how do you how does a healthy family or a healthy group of friends deal with issues like this you know how do organisms deal with issues like this you can put pressure in various ways on an organism that is being anti-social you know or an individual in your family a brother that's being anti-violent or anti-social there's various ways you can try to pressure them maybe to behave more socially appropriately or uh so that so that would be part of the the conceptualizing and the design of these frameworks is how do you deal with how do you deal with problems when when some club is not is you know misbehaving or some individual and a club is misbehaving there ought to be that ought to be well thought out so you have a procedure that you follow when things like that happen right but but just off the top of my head there would be many things you might do say a club was you know somehow turned rogue like a cancer in this community well you can you can decide not to trade with it you can all the other clubs can say you know what this is not this is not part of our larger community this isn't right this doesn't work and we're gonna exile you you know i mean various things you might do but all of these have to be given thought of how do you deal with the situations that arise and uh you know there there could well be reasonable ways that the communication through you know through other means that you could start to harmonize the whole if it's in a state of disharmony thanks i mean obviously obviously the human body does that right i mean the human there's trillions of cells in the human body and somehow they basically all work together and on a rare occasion some cells decide not to the the and we might call that a cancer so you know that cancers can happen and how do you then put you know the question becomes how do you prevent cancer from happening and obviously what i would say obviously one way is keep the organism healthy you know keep it keep it all healthy and that helps um keep communications open you know that that helps and then if cancer still arises then you know how do you then we have to have a way to deal with it thanks john so blue and then stephen so john i just want to say thanks i've really enjoyed all of your streams thus far and this is like a huge problem that um you've attempted to tackle here and so i don't want to like um approach it in in a way like it seems like i'm a naysayer or like i'm boohooing i just there's some extra components here that i think um you know maybe you have thought about or have approached it in a very delicate way so the concentration of wealth and power power i i feel in this country and this is definitely personal opinion or maybe in the world it is it like stems from some kind of spiritual bankruptcy right and and we've always had authority and like you talked about authority and societal cognition in the last paper and you talked about you know how authority comes from either voluntary deference or coercion right and previously from you know the beginning of written record until now and maybe even still now that voluntary indeference and coercion both can come from religion and religious groups and we've seen the god gene right and so i know that like you've kind of sidestepped and produced a very agnostic platform and and i mean i know that you also did talk about like the interconnectedness of all things and kind of um bringing that forward in an education way and not a religious way and i i definitely appreciate that but do you think that they're it's possible that we're hardwired you know we have this god gene right like that we're hardwired to need something beyond and that that if we have that something beyond maybe the concentration of wealth and power if we all gave you know if we all did tithing 10 of all of our salary i mean i think that that would equal out the world just that right so so i mean is it possible that religion is absent and maybe plays the more central role in society than you've allowed for here yeah i've not talked about religion much at all in any of the papers nor in the series these discussions and i don't i don't really want to um i i think this program can go and continue it can move forward and reach its aims fine without without focusing on the topic of religion now you know because people people belong to all kinds of organizations including religious organizations they people get meaning from all kinds of organizations and associations so good you know fine that's fine the the topic here is how does a community cognate how does it make decisions how can we create systems that favor beneficial and healthy cognition right so those two things can exist at the same time you know people can belong to religions and um and and participate in healthier systems at the same time um as far as meaning goes uh you know people find some people find meaning in religion um but in the uh i think it's useful for this project to um consider the the some of the ideas that i had in my first paper world view because those ideas of like how we are all interconnected how we are like trans really in in some sense transcendent ideas that that have scientific you know a rationale behind them right um that's useful to consider those points in the worldview paper because that at least in my in my thinking those concepts are are central to even coming up with the the uh concept of purpose so what is the if we're if we're together in a society what are we doing what is our purpose like if we don't have a purpose why are we even associating with each other you know just randomly or like just to take advantage of each other like why are we even in a community together you know like whether it's a large community or a small so i'm suggesting in the particularly in the first paper that that if we come together with a purpose you know and that purpose is to thrive now and in the future then we can do something about that we can you know we can create systems that allow us to fulfill that purpose that are true that are true to purpose the technology in a sense that's that functions the way it's supposed to function um and the the concept of purpose comes from this larger world view of who what is a human what is an individual how how are we all related to all of life um you know that's at least my mind that's kind of where where the idea of purpose comes from like you know that we're actually together as a society we're an organism and an organism is connected to all other organisms and all of life is connected all the way up to the biosphere and our purpose and include all of that so you could call that a trans you know a transcendent perspective perhaps and that might be i'm i'm hoping that might be meaningful folks who might want to be involved in this kind of project because uh you know if it is there's scientific reasons for all of that you know there's a rationale for all of that that we can explore all of that scientifically and otherwise you know uh so so yes people belong to religions and they can continue to belong to religions and and and from my perspective that that you know those organizations will will continue to exist as long as they're useful just like any organization just like any association it should continue exist as long as it's useful it's not going to happen in the club setting you know some people might be religiously oriented that's fine and but that's not really the you know the the the club focus the r d project the the design of new systems the engagement of people in a societal cognition framework that can all happen whether someone is religious or not i would say so so just as i said did i sidestep it well enough there well can i just ask a quick follow-up so so definitely in the um societal cognition i mean i just think that um we have to factor in the role that religion plays in the societal cognition because we give our we do voluntary deference and coercion through religion and and i think that it's something that if people ascribe to a religious practice that we allow for that and examine what role that will have on societal cognition at large like whether they're part of a religious organization or not or whatever oh sure those those are fine questions does does say you know does religion a in some ways impact the capacity of these systems to function in the way we're thinking they could function that's a fair question that's a scientific question or perhaps examining the the framework through the lens of each religion would be maybe useful and that would be useful also how that would be part of the question of like you know is there some impact that this that religion a is having on this system and uh if so let's talk about what how do how does this group see the world how does that how does that you know how do what's our differences of the way we see the world you know those are all fine questions and all of those questions can be addressed in a kind of a scientific manner when um when scott mentioned that there's the buyer in the cellar but also there's like the umbrella of the trust in the sovereign in the fiat you know fiat lux uh is how the book begins in it's a lot like a marriage partnership where there is a third party and that third party is the context and various religions have different ways to talk about that third entity but it's so interesting how when we're dealing with identity and communication trust it is like we need to have a higher level scaffold now can nature provide that can so-called science today or in the future provide that that's what we're going to discover and discuss so um stephen and then lee and then dave yeah this this idea of the environment and how we connect in to say a shared epistemic niche a way of understanding and kristen talks about that in some of his interviews around you know even religion can help us have a shared i think it's an epistemic framework or an epistemic niche for how we can align and that then ties into you know how do we treat hidden states if we're going to have active inference so as well as dynamical system in the brain and body you know the niche and where we are in the niche becomes a big part of that and of course what we've seen over the last 200 years is there's more and more states that are being fixed in the environment the things that we say we know or mechanical objects which will take over and get rid uncertainty markets will supposedly deal with all of that all these different um and it leaves less of a role for trying to attune to hidden states beyond so that that may be um one of those questions around you know money being a classic case of that you know how much do we trust in the econom economics and that will solve you know just vote for the best economics and it will deal with everything um but where is that is that really a deontic queue like fristen talks about deontic queues are accusing the environment like traffic lights that we just build all the traffic lights and then we just work to do what they tell us but that's where that violence can come in that i think was also mentioned is because deontic cues will always win if they're very strong over active influence in a community because active influence in the community needs integration it needs time to actually listen to the landscape and if you can have deontic cues that say well just you know that's not needed you kill a lot of that so in some way ways that might be relevant here is showing the benefit of epistemic niches in some ways these conversations are related to some of the points that blue is picking up on so so you know um if you try to to talk a little bit about about this so religion is a part of culture you know you can think of that as religious groups as a different cultural groups kind of social cultural groups right um you know culture is really useful uh no cultural norms are really useful can be useful they can draw our attention the things that in the world that are most important and help us to avoid danger and you know healthy culture is is a part of cognition it's a way to offload my cognitive burden onto the you know history of this culture that has learned as it has gone through life what works and what doesn't work you know so um so so all you know it is good that that say we had a large club with multiple multiple subsets of cultural subsets in it the the the again the purpose of the is societal cognition right and we we should have ways to measure if that's functional or dysfunctional both in are we achieving a quality of high quality of life are we you know are we are we is our uncertainty about the future high or low or you know how are we doing so so part of that is is engaging with cultural groups of subsets you know within a within a club or within a community right the idea of all of this what you know like what is the idea of cognition well cognition requires communication there's no that cells in the brain don't if there's not communication that's going back and forth rich rich communication right so we can decide as a as a as a group you know we can we can we can we can study different kind of designs for systems we can measure their success we can you know we can we can apply a kind of a scientific method to come up with good ways to ensure that we're we can all thrive but part of that is also uh allowing for and providing for and facilitating the communication between this group associations that are subsets of any club or any society right i mean idea has to be communication and in in a cooperative sense right or else there's no cognition to begin with you know then you just have rules and it's who god knows who made those rules in the first place you know so we're just all we're doing is like just just look at how cognition works and let's be honest about it let's be transparent about it let's you know let's just peel it peel back the covers and see how social cognition works and see how cognition works in biology and learn what we can learn from it and and uh let's bring the richness of that out and make it all transparent and fair so that allows for subgroups to have cultural differences that a lot that's important because cultural you cultural subgroups will have a different perspective on how to do things or what's important or where we should focus our attention you know that's all good right as long as as long as it's set up such that we're all able to openly communicate and tell our story and tell our narrative whether that's as an individual or whether that's as a cultural subgroup you know there's a narrative to express to there's concerns there's desires there's needs you know again humans are very complex organisms we have very complex and all of those you know all the core needs need to be fulfilled in whatever healthy society we build the need for association the need for love the need for helping each other like we you know we want to help each other we have to address all of that and it just has to be open and transparent and openly discussed about what we're doing and how this works and how people you know how how these tools and approaches facilitate cooperation and cognition or hinder it that's the experiment thanks john lee and then anyone else with a question uh yeah i think this is a a related question um i'm just going to read out just one thing from your paper actually because um so you in this paragraph you uh you say um in a second-order approach to active imprint um if a society understands itself as a cognitive organism uh if it understands its cognitive process as reducing uncertainty for the purpose of achieving and sustainably maintaining vitality and understand its social systems as a cognitive architecture then given those beliefs what kind of societal systems might it develop and employ um so that got me thinking is that are those beliefs prerequisite then for everybody in the club does it need to be that level of shared understanding or can it be just held within a kind of a subsystem you know is it is that something that perhaps the the scientists who are doing the simulation etc right right that's a good question now um you i think we should uh allow that if this r d program were to go forward that you know it might start small and focused and there might be a focused group involved of those people who are interested in this kind of thing right and if it started to work and you know it proved to be beneficial to communities then it might grow and other people would get involved so we can think of this whole thing as progression i think that's fair that's the way it would naturally play out and uh you know if the scientific community is involved in this and if the scientific community is involved in creating and helping to you know co-designing new systems well then there's going to certainly be a focus on the scientific viewpoint in order to do that communities obviously would have their input into the r d program and you know it would between the two between the two it would move forward but i would guess though that early participants and early efforts would be uh focused you know would more or less align with what i have in the worldview paper or else they why would they be interested in participating in the first place you know like they would maybe only be interested because they'll they can see some alignment with the goals of the program and the way the program is oriented and what it's trying to achieve right so quite likely the early participants in this whole thing whether they're scientists or community members might be kind of more or less aligned with ideas in the world of your paper and and especially the purpose of you know the purpose of a club or the purpose of societal cognition now suppose down the road you know 20 years from now i live in los angeles and they're you know like i don't i don't care at all about any of that stuff i just want to live my life and you know make sure my i have a roof over my head and stuff but this club is up and going and is doing well and they're all there you know they're actually doing well my neighbor is a part of it and his life is improved in some way so like i don't really i don't really you know like i don't i don't care about any of the purpose stuff too much but i like what's happening you know like they got a nice scene going on there that's beneficial so i'm gonna join anyway you and it's so to a degree the the culture of it would change it would evolve but hopefully that would be it would start with a certain kind of it would start in a healthy functional way you know aligned with the purpose of the whole project and then it would evolve the way it evolves you know i would i would guess that if it's beneficial those beneficial ideas would continue to be the you know at its core somehow but how it exactly it evolves you know as as decades goes on and as you know as is 100 years from now it's maybe maybe you know we if this exists 100 years from now and if it's spread widely maybe we're using very different language to talk about it and we see it differently perhaps thanks john um now as the author it is up to you would you like to shift the second slide for today these are awesome questions thanks everyone in the chat and for participating because it's it's a conversation and john's uh initial inroads are tripartite and very provocative and we're seeing that play out and so it's a great beginning to a discussion even though it's our dot 2 video so john where would you like to take it for the last second uh yeah obviously there's a lot of ideas in this whole series there's just a ton of ideas uh both both uh superficial and deep and if this goes forward it's gonna take lot of discussion and it's gonna take a lot of people involved to bring different perspectives to kind of make bring this whole concept to fruition right so we're just scratching the surface that's all we're doing we're just putting out some of the big ideas and then you know there's a ton of work that has to happen to make this real so uh can you go to uh four number four group four slides maybe uh um maybe uh slide seven six if uh seven integrated components on slide six six yeah yeah seven integrated components yeah awesome okay so uh maybe in the last little remaining time we'll just touch on this lita framework the simulation model that i put out it's really the the idea of the leader more than just an economic system it is actually the idea of it that is put out in the in uh in my book economic direct democracy um talks about the whole picture the larger picture but there there's as you can see from the slide there's seven components to that and and they really touch on all six overarching systems that i talk about in the series so economic governance legal analytical education and health it's all in there somewhere so the idea of the leader that's local economic direct democracy association the idea of the leader is an integrated framework of cognitive systems the simulation really focuses more just on the economic aspect because that was the eat really was just the easiest piece to pick up first um and uh the the the uh in particular it focuses on the what i call the token exchange system which is the first three in the list so there's a token is a is a local digital currency so um that the community creates and uses uh there's the token monetary system there's what i call the crowd-based financial system uh and there's a market system local markets people sell you know creating goods and sell them in local markets they also obviously sell distantly to people who are uh either in the local area but not a part of the club but also far distant that are not a part of the so those uh first three things constitute the token exchange system and we'll just maybe talk a little bit about the simulation and some of the results and how that works um maybe slide eight yeah yep so in the simulation there's five aggregate agents uh there's persons the cloud-based financial system local government and then the rest of the world which i call rest of counties the the data for the simulation the initial data was from the u.s census for eugene oregon relatively small community a small county population 100 000 or something and i used starting income values from the census data for eugene county from 2014 or something um this paper was published i believe in 2014 so i probably used a little earlier data to get things rolling okay so the the simulation just it really illustrates it doesn't it doesn't predict how the system would work it just illustrates my thinking about how a system could work and in particular i wanted to uh i wanted to design a system that would that could bootstrap itself up without large influx of cash or other things from outside so that was part of the idea how can a system start with very little and drove from there the second was i wanted to create an economic system for the for the club model where over time incomes became equal income income and wealth equalized the the thinking there is that money and and and you know the whole economic and financial aspects of that really is part of the cognitive community's cognitive system so we're using money as a very transparent voting tool right it's like votes information it's carrying information about people's desires and fears and everything um uh yeah so that's uh that's i wanted to show that using a stock flow consistent model agent-based stock stock flow consistent model i just wanted to show that the numbers add up correctly right like i'm not i'm not i'm not creating uh dollars out of nowhere there's like a there's a logical way this plays out and all the numbers add up there if there's a flow from one agent uh one of these major agents there's a you know there's a any flow from one agent goes somewhere it just doesn't disappear uh so uh very briefly the uh the lines in blue are the flow of tokens and dollars so um a community is creating tokens as it goes as it grows it creates more tokens the the concept is that the value of a token would be sort of it's sort of equivalent to the value of a dollar and it's made to be non-inflation inflationary so let's just just say for 2014 it was the value of a token is the value of a dollar purchasing power of a token is that of a dollar in 2014. and uh um the community would create tokens a little bit in the first year a little bit more of the second years more people get engaged there's more use for tokens there's more ways to and you know use them in in the local markets then there would be more tokens

tokens created and that and the and the volume of tokens in the local community would grow and salaries for those who were members of the club salaries would be paid both in tokens and in dollars so maybe in the first year you would just get a few tokens and then over time your share of tokens in your wages would go up and the same would be true even if you're just saying you you yourself own a business you know people you would sell your if you're a member of the community you would sell things to other members of the community and they would pay you in both dollars and tokens amusing dollar as the national currency so organizations just to look at the blue lines again organizations pay wages to persons the persons might be employed or unemployed i i also call unemployed not that there's some people who are unemployed or not in the workforce that's what an if niwh stands for and uh people who members who receive dollars and tokens as part of their wages as part of their income would naturally they would have to contribute a certain fraction to the crowd-based financial system and uh that's the blue line for contributions to the cbfs and the cn from the cbfs if i'm an organization or i'm an individual and i have some idea to start a business or i need i need loans for my business or i'm a non-profit and i want grants i can go to the cbfs and apply for a grant or or a subsidy or donation and once as a person once my once some of my currency is in the cdfs i more or less get to decide how that money is used like i want to support schools or i i like this idea of a new light bulb i like that i like that business idea that this person came up with i'd like to give them a subsidy or i'd like to give them a loan or a donation or something like that again this is all very transparent and and again as time goes on incomes uh become more equal so that and then because incomes become more equal everyone has equal amount of of power in the cbfs to direct how the community spends its money the green lines in this figure are dollars and obviously you have to pay taxes and dollars the the government's not going to accept tokens so um uh you paid all taxes and dollars the government also provides some income support for people who are unemployed natural normal the government also provides some grants and contracts and subsidies to organizations that's quite normal and organ organizations that our members also have to deal with the outside world they have to sell their goods and and exchange uh goods and services using dollars so those are the green lines uh so that's how the basic system works and then i and then as this this the system evolves over a period of 28 years in the simulation and i'll show you what the results are as the as as everything goes round and round we're not going to have we only have a few minutes left so i'm going to kind of have to get just get to the point here there's some assumptions uh on the next slide what john can i just ask um which of these agents are active inference agents or where does active inference come into play or could come into play with later versions of this simulation um okay so again the the the um the leader framework is an integrated system that has a governance component if the governance component has a legislative component and you know all the major six overarching systems are addressed in i'm only simulating the economic part so if you were in a real club and a real club looked anything like the leader there would be a governance system that you would be you know participating in some kind of collaborative governance system to help decide you know what should happen in the community that'd be the legislative system legal system and so on uh education system and everything so that's one way that active imprints would be involved just in the process of decision making under uncertainty right and in the in this particular simulation you know obviously businesses are having to make decisions and choose choose what they do with their resources so that's another way that cognition is happening and then maybe the main in this economic picture maybe the main agent is the cbfs that's a that's a financial system where everyone is able to participate in a transparent and democratic way to fund those things that they think are useful to their community that create the kind of jobs that they want to create to you know do as a community do we want lots of non-profit jobs do we want for-profit jobs do we want jobs in the manufacturing center do we manufacturing sector do we want you know like do we want to support this kind of business do we want to support that kind of business these are all decisions that the community makes through the cdfs and then you know if if the community decides like yeah we want this bakery this would be great or this bookstore or this farm or whatever it is then the community can you know associate within the cbfs and say okay there's a thousand of us we like this idea for this farm here's funding for it so you know go for so active inference particularly would be happening in the cpfs yeah steven a question and then anyone else with a question yeah i suppose in a way you've seen a mix here between like in the in the red boxes you could have active inference like the dynamical non-linear systems of inference and then you've got this message passing which to some extent is then the aggregated system the sort of steady equilibrium i the pile of money or the spreadsheet and a message passing where data goes from one to another but then within that there may have to be another process of inference that happens assuming people have any agency and it's not just all automated so it's like to me that's probably where you've got system dynamics in that diagram and then you've got these inside that non-linear dynamical systems i.e the inference processes embedded inside yeah there obviously is a lot going on there that we're not going to have time to talk about because we have like five minutes left or ten minutes left but but just to give you one example of suppose that a a a group in the community wants to create some kind of business and they want to produce some kind of product right so they come to the cpfs and say hey we want to make this new thing well a lot of information has to go back and forth right this is an information rich system so i wanna if i'm gonna if i'm gonna support this with my resources in the cbfs i wanna know where you get what you know where are you getting the raw materials how is that impacting the environment how is your product going to improve or harm my our sense of community or any one of the metrics that we're measuring how is that gonna is there any pollution how are you gonna deal with your waste how much energy do you need uh do we have that that level of energy like electricity or whatever as a community do we have that resource about i mean there's an enormous amount of discussion and information has to flow between these different parties in order to make the kind of decisions that you know intelligent decisions that you would want to make in the cbfs okay so um in the few minutes that we remaining if you could turn to slide 15 please all right go for it so these are some of the results um uh um and i should say that um some of these that right at the get-go as the community is forming as a club is forming you know they would be choosing their trajectory they would be and and this upper left-hand corner is an example of that they would be choosing that they want income to rise to a certain level over time that they want participation to happen in this club at a certain rate over time they want to uh they want in particular the lower lower left there they want the income target to rise at a certain rate over time and um and obviously you can't ju you you know like you can't just flood a community with money on day one and expect that to work because there's no the same with uh say with digital currency because there's no no place to spend it like there's you know it's a tool for voting and if there's no place to vote the tool is useless right so that's partly why things grow slowly over time so that there becomes outlets for using the currency it you know and to create the kind of economy and and economic system that the community wants some of these things would be decided ahead of time like here's what we want as a trajectory here's what we're here's what we're doing as a community using the system this is where we're going this is what we're going to do this is how it works and if we all cooperate in this way that's exactly how it's going to happen you know that's that's the concept is so it's not just here's a system use it however you want let's see how it goes it's really more you know we we're a community that we have a goal and here's how we intend to reach our goal we want say for example we want incomes to rise to a certain level over time and this is you know can this system do and if the answer is yes well then here's how we're going to do it we're going to increase it a certain amount every year and it's all going to work out and and so in that sense some of this should be no there should be no surprises for the community as they do a field trial it should be working exactly the way they intended it to work bringing them where they want to go for example when you drive a car on a trip you you know where you want to go if you're halfway there if it's a two-day trip and on day one you're halfway there then you know you're moving in the right direction it's all going just where you intended to go okay so um on this slide i may be on the lower right hand corner uh this is what happens over 30 years as the simulation moves forward uh the this is showing average family income in the in the membership it starts at a fairly low level this is from the census data this is where they start on your zero at your zero and every year the average family in income increases essentially the leader is paying people to join either because the because the income target is rising and my income is go my wages are going up as the income target goes up or i actually have make a lot of money but i'm joining anyway just to get that bonus uh you know whatever it is a thousand dollars a year in tokens or whatever that bonus is so people are joining because it is it in their economic best interest to join and as it goes on the income target rises and then family income you can see the blue line there family income average family income more than doubles over the 30-year period so uh the average family between tokens and dollars the average family income is over it's around 110 000 annually as as time goes now i might note that the dollar aspect obviously the community needs to bring in dollars in order for this to happen because some of the income is paid and some of the wages are paid in dollars and some are paid in so part of what this community is doing is they're adjusting their outflows they're adjusting their transactions with the rest of the world such that they're keeping dollars in circulation longer than would other but otherwise be natural so for example there's a like a buy local program and there's ways to support uh local local businesses and keep money keep dollars circulating in their local economy longer and in the end what happens is at this high income level what what actually happens the community is is is increasing the the number of dollars in circulation locally up to the point where their average dollar income is the national in average national dollar income so you like today in today's world the incomes are very very uh unequal there's a few extremely wealthy people and most everyone else is not making very much but there is an average dollar you know average dollar income over the country and what the leader is doing over time is is kind of pulling dollars into their local system such that they're just reaching the average overtime and then they so they're reaching the average average dollar income which is about 70 000 or so dollars and then they're also have on top of that uh income in tokens which the community is creating so between the tokens and dollars eventually they get to the point where where they're having about a family income of about a hundred and ten thousand thousand uh so somebody had said earlier that you know maybe it was you daniel that i'm both i'm both saying that money is a different animal in this system and motivations are different and in the next breath i'm saying oh this is a motivation you could you can make more money so like both of those things are true at the same time and um and there's a transition period where we go from here to there right i mean this whole thing just doesn't happen overnight if we want to transform society it takes time and uh so in this transition period we might we might be thinking of money and income in both ways like in the more traditional sense of i need money to pay my rent and then in the new sense of money is a voting tool and uh we all need to have more or less equal income so that we have more less equal voting power in this community now what happens happens after 30 years or 50 years you know maybe our understanding of all of this deepens and widens and um you know maybe the the current concepts of money and economy fall away in exchange for this new new uh focus on cognition and well-being and uh value of community depending on how well it's thriving is is this is the code is the code or the formalism available because people might tweak a parameter and get a totally different response they might include another variable or another function get a totally different outcome so how can we actually version this in an open source way so that we can be working together on something that's transparent yeah yeah the code was made of open source but made available publicly back in 2014 um and i also there's a interactive simulation on the website it needs a little artwork but it functions so you can go to the website principalsocietiesproject.org and you to some degree you can type in some different numbers and see how that all works out so you can play with it a little bit on the website uh but beyond that um this simulation is really quite simple and if i were to recode it i might record it in a different way today you know that i did back in 2014 but this the simulation is really quite simple it needs this is just a first glance of a conceptual glance of how this could work and it needs all kinds of work there's all kinds of ways that we could expand that you know there's a lot of questions that need to be answered and we could expand the simulation to answer questions better or address more questions so uh so you know you can download the code that i put up in 2014 but you know ideally i think we would move forward and develop you know kind of a more functional framework for this that we could try out different approaches and different ways to do it um uh maybe on the slide 16 in these last few minutes this is maybe the one of the kicker slides you can see on the right the histogram of this is starting distribution of income in the county and then below is the income distribution at the end of the 30-year period and you can see practically everybody is making this 110 000 in currency per year on the left you can see how the unemployment rate changes it starts at about seven percent or so uh in the county and drops down to essentially full employment on the next slide slide 17 on the top left is the curve of how much money is in circulation in the cbfs over time and you can see that about 6 billion in currency circulates annually at the end of 30 years in the cbfs so again this is a highly transparent uh deeply democratic and collaborative way to decide how money should be spent in the and imagine and this is a very very small community over like 100 000 people or something like that so this is six billion dollars for a hundred thousand people to decide how they want to you know what they want for their community what kind of jobs they want what kind of what kind of uh you know how many non-profits they want versus for-profits how what they want like you know that's a lot of money and with that much money a community would have enormous flexibility and capacity to make the kinds of decisions that it would want to make about how how it lives its life maybe that's the that's the you know that's maybe the main slides the main results essentially income more than doubles enormous amount of money circulates annually in the cbfs and the community gets decide gets to decide in a very transparent and deeply democratic way how that money is spent so maybe we'll end there since we're out of time but there's obviously a lot more details in the simulation paper and anyone can read that to pick up more thanks so much john so we'll chill for a few more minutes people can drop off whenever they want to but maybe if anyone who wants to raise their hand uh and give a final thought but it's awesome that the code is posted maybe you can add that in a youtube comment because it'd be awesome to see maybe where this even plays nicely with the active inference models that we've been exploring in these recent discussions so any um one wants to raise their hand give a few more seconds for that also one note about um communication and stephen was talking about like the agents active inference agents and message passing between agents also we know that message passing is a mechanism by which active inference can be computed within an agent there's a lot of cool papers on that and it reminded me of um some work that i had done in the eu social insects in the ants which investigated how physiological mechanisms that were inside of a solitary insect's body like related to the regulation of foraging over evolutionary time in the colony context become rewired to be playing out across multiple bodies so the regulation of foraging for a solitary insect is all on board the fly but the regulation of foraging for the ant colony is something that's distributed and it got there through evolution and hundreds of millions of years of it working so yes there are failures just because one is seeking a goal doesn't mean that they reach the goal most ant species of course are extinct but it does point to this way where the kinds of connections that are inside of an agent can be of one kind and then the communications between agents can be we see that with neurons with ants with societies and by having a scale free framework like active inference we could rethink how communication happens inside and outside so that we can take some agency ourselves in designing our niche in our evolutionary transitions upcoming so dave and then lee um i see some um data from the actual physical economy in eugene is there um any interest in setting up um leon tf matrices and running an actual physical simulation of the economy so you don't just swap everything into into money but actually use physical activity and see whether you're exceeding break even over a period of decades i'm not if you mean by physical you mean like an actual club yes that's where all of this is going you know like in in in whatever would take six years ten years whenever this is ready yes let's do a real let's do a field trial in a real club and see how this works but if you're talking about more like uh in silico sure all i'm looking at here is just some simple metrics about about this change exchange of currency but there's all kinds of things that have to be exchanged and considered there's energy there's wastes there's you know pollutants there's uh there's raw materials there's uh human resources there's like all of these things all of the components could become part of a simulation model and should become part of simulation models over so again this is a very very simple simulation and there's tremendous opportunities to expand it and make it more meaningful thanks lee i am sorry um i'm gonna try and get two questions for the price of one so my first question is um given that there's you know you see this kind of prelude phase um what what would you see as uh priorities initial priorities for for research then to get this really moving yeah i mean good question and and and i wish i had a uh you know a clear path to let's do a b c and d and then it's all going to work out but i don't so we have to decide that together really i mean i could i i could come up with a list of ideas that we could all discuss together you know for just one is the these kinds of simulation models could be expanded but there's other things too for example it would be really interesting to just do a survey either of the general population in you know in one country or maybe multiple countries or do a simulation of the global in the global science community and just say to scientists you know if this kind of r d program were were available or were you know it was activated and there was funding available would you be interested in participating in this you know like it would be really interesting to know is there support for this in the scientific community and in the public does the public is the public interested in fundamentally new systems you know those would be interesting questions there could be media some you know somebody should make a documentary about the concept of society as a cognitive organism what we could become you know like they could be art projects there could be media projects it could be literature projects that could be simulations there could be surveys there could be assessments of say you know choose some community somewhere maybe a smallish and assess what is their uh how do we apply metrics to their current fitness what kind of metrics they could be work on metrics so they could be work on applying metrics to a to an existing community just as a just to get a background of like where are we now and and how would we collect this information there could be that would you know there could be work on computer models to be like you know if we had 50 metrics how do we combine them into some something meaningful for uh you know for making decisions you know like so or on the topic of tools how what kind of tool would allow 10 000 people to richly communicate and make decisions together and and how can i give how can i tell my story a thousand people or ten thousand people how can i can tell how can i tell my story my narrative my understanding of things in some rich way such that it is computed and used by the whole in a in a meaningful way there's also you know a few ideas you know the right yeah thanks for the answer and live stream is a new affordance it allows those to speak but with a hundred thousand people there isn't enough hours in the life to linearly hear from everyone and so when we want to speak and we want to hear and listen active listening passive speaking all these things how are we going to make that happen at scale so awesome questions you're raising steven and anyone else and then we'll close out yeah i think the scale question keeps coming up right that's the start to realize that's that's everywhere but often is is not talked about um one thing just uh sort of to finish off was there's the inferences about how to act in the world which but there's also the ways to get an insight into the influences that we're making so in some ways i always think this is giving a way to look more deeply um and go beyond you know system dynamics which is used quite a lot in population work but to go into this kind of um way to sort of say okay we're not necessarily creating these clubs to tell you what to do and to replace other government structures but to give other ways to get insight into what is going on in the inference process here yeah excellent excellent so this is um you know potential this whole concept is potentially useful for making actually making decisions decisions for the future as a by a group but also for individuals and a group to understand how are we how do we cognitive how are we cogniting how can we do how could we cognate better or how do we cognate how does how do we actually function you know like all of these are questions and all of these are open to assessment and you know like i could imagine thousands if this went forward there could be thousands of papers on all kinds of topics not just about cognition but you know that would include cognition about how does it actually work how do humans how do humans associate and cognite together what a time it will be so thank you john it was great to have you on and the six appearances were just the tip of the iceberg so we look forward to continuing the discussion with you with seeing you in our tools organizational unit where we're really inspired to be working towards developing some of those things that you're suggesting that there's a need for so thanks everyone for joining live and for watching and for the awesome questions yvonne natalia etc so we'll see you in the coming weeks with future papers and future discussions so see you later thanks much bye